

GRAFT: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

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Atom-to-atom mapping for reaction analysis requires retaining alternative correspondences without enumerating redundant atom permutations. We introduce GRAFT, a graph-based method that grows matched fragments on weighted reactant and product graphs, represents compatible placements through correlated symmetry actions, and branches on distinct saturated fragment placements. The default pipeline uses one seed ordering per cut and the cut sweep. Completed searches are deduplicated by unordered matched fragment pairs while preserving their internal mapping alternatives. A symbolic decoder extracts distinct signed bond-event patterns directly from these families, separating the tolerance used for matching from the thresholds used to interpret bond changes. On all 1,851 Golden benchmark records, bidirectional sweep search recovers an annotated correspondence among its retained alternatives in 1,834 cases (99.08%) with one seed ordering per cut (default) and 1,840 (99.41%) with ten. All GRAFT reference-recovery verdicts are resolved. The strongest evaluated prior-method alternative-coverage baseline, the bidirectional SLAP sweep, recovers 97.03%. On the matched 1,821-reaction timing cohort, the one-seed GRAFT search stage and SLAP sweep average 2.52 and 4.82 CPU seconds per reaction, respectively. On 140 coordinate/Wiberg-bond-order reaction pairs, GRAFT with a 2,000-branch cap reaches the same minimum event count as the SLAP sweep in 136 reactions and fewer events in four. Where their minima agree, it retains 158 of 164 comparator minimum-event patterns. At the default cap of 100, 138 reactions return complete mappings and two return none. This bond-event comparison uses unverified mappings and does not measure mapping accuracy. GRAFT provides an inspectable representation of mapping and bond-event alternatives for subsequent chemical assessment.

Date: September 12, 2026

Status: Preprint



1 Introduction

Atom-to-atom mapping assigns corresponding atoms across a reaction. It underlies reaction-center identification, template extraction, and the preparation of endpoint structures for mechanistic calculations. A reaction can admit several plausible correspondences, while many different atom-index assignments are chemically equivalent. An algorithm intended to retain alternatives must therefore resolve two questions: which structural choices deserve separate candidates, and which permutations can be represented together?

Existing approaches use learned correspondence signals or graph-based optimization. RXNMapper derives mappings from attention in a reaction language model (11); LocalMapper learns from chemist-labeled reactions (2); and SLAPMapper combines iterative label refinement with sequential linear assignment (7). The Golden benchmark provides a common set of annotated reactions for assessing mapping recovery (8). For applications requiring alternatives, recovering the annotation somewhere in a returned family is a different objective from producing one correct top-ranked mapping.

Weighted endpoint graphs introduce a further distinction. A permissive matching tolerance can preserve a fragment despite changes in its bond orders, but the corresponding atom shuffles need not preserve the bond events assigned at a finer threshold. Replacing a compressed family by one representative can therefore hide distinct transformations. Likewise, greedily growing one fragment can prevent a neighboring fragment from reaching an alternative assignment.

GRAFT addresses these issues by branching over structural fragment choices while keeping symmetry-related atom bijections compressed (Figure 1). Continuous growth and alternative placements determine the structural choices. Default GRAFT uses one seed ordering per cut and the cut sweep. We introduce a cut-sweep strategy that changes the search conditions to counteract greedy fragment commitment; it is part of default GRAFT, and we also apply it to SLAP to test its benefit beyond our fragment search. Its final representation groups unordered matched fragment pairs and retains the allowed mapping alternatives within each group. A constraint-based decoder projects these families onto distinct bond-event patterns. The method searches for structurally compatible alternatives without requiring the reference transformation to minimize the number of bond changes.

2 Method

Algorithm 1 connects the fragment-level operations in Figure 1. A live branch stores the fragments already placed, their atom assignments, and the compressed choices still allowed by those decisions. Growing and placing the next fragment conditions every later continuation. The matching routine is treated as an operation here; its compatibility tests and symmetry representation are defined below. For bidirectional search, the search phase is applied in each orientation, retaining the orientation of each returned family.

The forks in the search phase represent different fragment placements, not one branch for each symmetry-related bijection. The cut sweep changes the conditions of this search; a single mask can expose a mapping with several bond events. Event decoding is a separate operation on the retained output. Search caps are recorded as truncation, not treated as proof that no alternatives exist.

2.1 Weighted matching and continuous growth

Let V_R, V_P be the reactant and product atom sets, Z_R, Z_P their element labels, and W_R, W_P their symmetric bond-weight matrices. A partial mapping $m : D \subseteq V_R \rightarrow V_P$ is injective and element compatible. A mapping assigning every source atom is bijective when the endpoint element counts agree. Explicit hydrogen atoms participate in the search. Molecular coordinates can accompany the graphs but do not determine the growth decisions.

The active source graph contains edges with weight at least b . Starting from a seed atom, GRAFT grows a connected source fragment F while carrying multiple compatible product placements. A frontier queue prioritizes stronger source edges. To add an unassigned source atom u with product image v , a placement

Algorithm 1 GRAFT: from fragment growth to alternative bond-event candidates

Require: Oriented endpoint graphs, seed orders, optional anchors, search budgets

Ensure: Compressed final branches; optionally, distinct bond-event candidates

```
1:  $\mathcal{S} \leftarrow \emptyset$  {collected terminal branches}
2: for all conditions: uncut source graph and each single-edge source mask do
3:   for all requested seed orders under this condition do
4:     Initialize live branches with the supplied anchors
5:     repeat
6:       for all seeds in the current order do
7:         for all live branches eligible to grow from this seed do
8:           Grow a fragment to saturation, conditioned on this branch {Fig. 1a}
9:           if no admissible fragment placement is found then
10:            Retain this branch for later seeds
11:           else
12:            Fork one continuation per distinct structural fragment placement
13:            Commit each placement; retain its compressed symmetry and constraints
14:            Admit the continuations within the branch budget
15:           end if
16:         end for
17:       end for
18:     until all branches are finished or a full pass makes no progress
19:     Add terminal branches and their search-stop status to  $\mathcal{S}$ 
20:   end for
21: end for
22: Group final branches by unordered fragment pairs; retain their family union
23: Expose compressed branches with constraints and provenance
24: if bond-event candidates are requested as separate post-processing then
25:   Decode distinct event classes from complete families using the original matrices
26:   Report witnesses, supporting families, and completeness status {Fig. 1c}
27: end if
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must satisfy

$$Z_R(u) = Z_P(v), \quad v \notin m(D), \quad W_P(v, m(r)) \geq b, \quad |W_R(u, r) - W_P(v, m(r))| \leq \tau_{\text{iso}} \quad (1)$$

for every active, nonrelaxed source edge (u, r) into F . Injectivity is enforced against all current assignments in D , including established fragments. All relevant fragment bonds are checked, rather than only the frontier edge that proposed the extension. Source nonedges do not impose negative matching constraints, so additional product bonds remain possible.

An accepted extension enlarges the fragment and filters its compatible placements. Growth continues even if only one placement remains. An unsupported edge is deferred, and other frontier edges can still be explored. When growth reaches an established fragment, ordinary continuation respects its existing assignments. A fragment saturates when no further admissible extension remains; structurally distinct surviving placements then generate separate continuations for the remaining atoms.

This is a greedy search along each trajectory: an accepted extension can remove placements that would support a different fragment boundary. GRAFT broadens coverage through cut sweeps and seed orderings.

2.2 Cut sweep to vary greedy fragment boundaries

Our cut-sweep strategy reruns growth under alternative source-edge conditions. In addition to the uncut graph $G_R = (V_R, E_R)$, it searches a graph with one eligible edge e removed, $G_R^{(e)} = (V_R, E_R \setminus \{e\})$, for each edge at or above the cut threshold. The evaluated cut threshold equals the active-graph floor $b = 0.2$, and eligible edges include bonds to H. Each subrun retains the atom set and the original endpoint weight matrices.

Removing e changes both growth and matching: the frontier cannot extend along that edge, and its bond-preservation test is omitted from Equation (1). An extension accepted in the uncut run may otherwise merge

atoms into a fragment and discard alternative placements. Under the cut condition, that growth route is unavailable; a fragment can stop at a different boundary, grow through another route, or leave atoms for a later seed. The missing preservation test also admits placements previously excluded by e . Thus the sweep counteracts greedy commitment by changing the conditions under which growth proceeds. It does not force the endpoints of e into different final fragments or require a bond-breaking event: they may reconnect through other source edges, and the final correspondence may preserve their bond.

GRAFT takes the union of the uncut and single-edge subruns before final branch deduplication. The default one-seed mode uses one seed ordering per subrun, so it already includes the full cut sweep. There are $|E_R| + 1$ subruns at the evaluated thresholds. A single-edge cut describes the search perturbation, not the number of bond events in a returned mapping. One removed constraint can redirect subsequent fragment growth and placements throughout the reaction, exposing an alternative with several bond formations, cleavages, or bond-order changes. GRAFT therefore need not enumerate combinations of final bond events to discover multi-event alternatives. Bond-event decoding uses the original matrices, so a search cut itself contributes no event or event-count penalty. Growth remains greedy within each conditioned subrun; the union broadens coverage without making the search exhaustive.

2.3 Automorphisms and correlated mapping families

The distinction between structural branches and explicit bijections is central to the design. Even one fragment family can contain combinatorially many assignments: k independently exchangeable methyl-hydrogen triplets admit $(3!)^k$ bijections. Expanding those automorphisms into separate mappings would multiply the work without adding fragment choices. GRAFT instead stores the permitted permutation actions and their constraints, and decodes event alternatives directly from this representation. Branch caps bound retained alternatives within each growth call; they do not guarantee a small global search. This avoids explicitly listing symmetry-related bijections within each retained family.

Compatible placements are stored using a representative assignment, unresolved assignment blocks, and permutation generators. During growth, graph canonicalization accounts for the current fragment, locked assignments, and boundary information. Each saved transition records its action generators together with conditioning and fragment-preservation constraints; their conjunction defines the allowed mapping relation. Canonical labeling and automorphism computation use nauty (9).

Matching tolerance and automorphism equivalence have different roles. The relation $|x - y| \leq \tau_{\text{iso}}$ is generally not transitive, so it does not by itself define a permutation group. To describe an intrinsic symmetry shared by a final fragment pair, we instead color product bonds by their responses to the fragment’s matching tests. For a product weight w and the set $\mathcal{W}$ of active reactant weights for the same element pair, the label records

$$\chi_{\mathcal{W}}(w) = (\mathbf{1}[w \geq b], (\mathbf{1}[w \geq b \wedge |w - v| \leq \tau_{\text{iso}}])_{v \in \mathcal{W}}). \quad (2)$$

An exact automorphism of this colored graph preserves every encoded matching test. This definition depends on the final fragment pair, not its growth order.

The saved mapping relation also contains the constraints imposed by its search decisions. We represent a flat family $\mathcal{F}_j$ by a seed mapping m_j , a sequence of allowed action groups $G_{j,1}, \dots, G_{j,k}$, and final preservation constraints C_j :

$$\mathcal{F}_j = \{a_k \circ \dots \circ a_1 \circ m_j : a_i \in G_{j,i}, C_j(a_k \circ \dots \circ a_1 \circ m_j)\}. \quad (3)$$

These actions encode correlated moves; atoms in a common orbit are not independently permutable. Disjoint actions commute and can be placed in a canonical order. The order of overlapping actions is retained. A sequence of groups is not generally equivalent to the group generated by their union.

The intrinsic automorphism of a fragment pair is shared metadata, not permission to enlarge its saved family. Multiple admissible seed mappings can occupy different intrinsic symmetry orbits, and different search constraints can restrict the allowed actions. GRAFT therefore retains these alternatives explicitly inside the compressed representation.

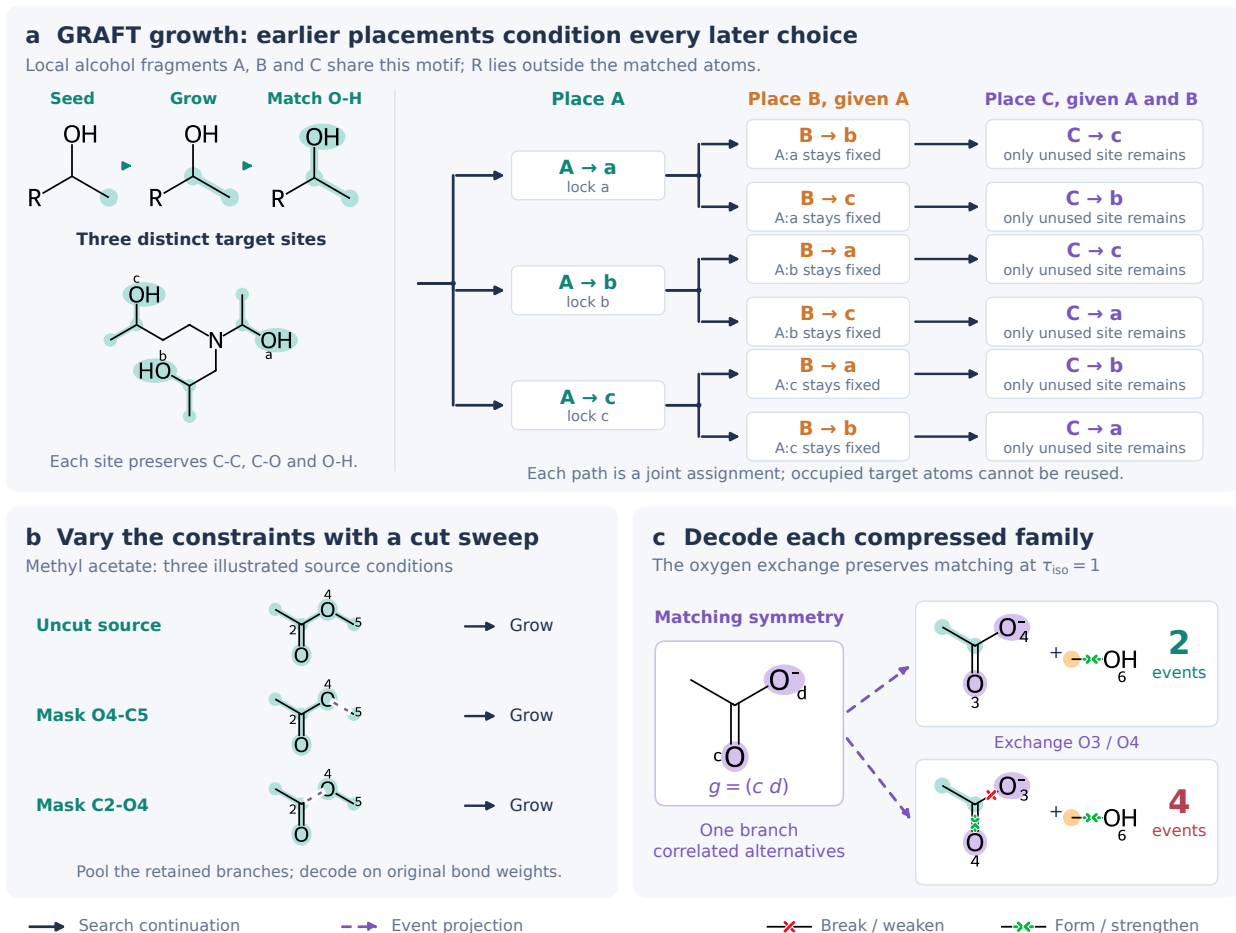


Figure 1 GRAFT: conditional fragment branching and compressed symmetry. (a) A local alcohol motif grows while retaining its O-H bond. Three sites in the depicted target have distinct molecular environments. For three source fragments sharing this motif, placing A fixes one site; B then branches over the two unused sites, and C takes the remaining site. The tree illustrates local conditioning and injectivity; R denotes atoms outside the matched subgraph, and this is not a complete reaction-search trajectory. All attached hydrogens in the illustrated local matches are checked. (b) The cut sweep searches the uncut source graph and graphs with one source edge masked. The illustrated ester masks change the growth and preservation constraints; their outputs are pooled. Masking an edge does not itself assign a bond event. (c) Exchanging the two acetate oxygen images preserves the fragment’s element and bond-weight matching tests at $\tau_{iso} = 1$, but changes two heavy-atom events into four at threshold 0.5. Neither acetate oxygen carries H. Solid black lines retain the endpoint bonds. Red $\times$ marks on the bonds indicate cleavage or weakening; paired green arrows pointing inward indicate formation or strengthening. Pastel colors track fragment membership or symmetry sites. Dashed arrows are event projections within a family, not additional search branches. Saved constraints govern allowed symmetry actions. These formal-bond-order event signatures illustrate GRAFT decoding, not distinct mechanisms or resonance-normalized transformations.

2.4 Unordered final branches

During search, history and boundary conditions matter because they determine valid continuation. At completion, the branch identity is instead the unordered collection of matched atom sets,

$$\mathcal{B} = \{(R_1, P_1), \dots, (R_q, P_q)\}, \quad P_i = m(R_i), \quad (4)$$

under a fixed matching policy. Sorting atoms within each set and then sorting the fragment pairs removes dependence on growth order and fragment labels. Growing A before B or B before A yields one branch when the final paired sets agree. The colors within each candidate in Figure 2 illustrate these matched fragment pairs.

Each branch stores the union of its flat families. Identical normalized family records are shared, while distinct constraints or action programs remain available. This transformation preserves the represented mapping union: it changes its grouping, not its admissible assignments. Provenance links retain the discovering paths, but the original search hierarchy is unnecessary for decoding. This conservative deduplication does not assert that the remaining families are mutually disjoint.

2.5 Decoding distinct bond-event patterns

For each complete mapping, define the signed change for an unordered source pair $\{u, v\}$ by

$$\Delta_{uv}(m) = W_P(m(u), m(v)) - W_R(u, v), \quad s_{uv}(m) = \begin{cases} +1, & \Delta_{uv}(m) \geq \delta_{uv}, \\ -1, & \Delta_{uv}(m) \leq -\delta_{uv}, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Positive events denote bond formation or strengthening; negative events denote cleavage or weakening. The event count is $E(m) = \sum_{u < v} \mathbf{1}[s_{uv}(m) \neq 0]$. The evaluation includes hydrogen atoms and all atom pairs. For the WBO collection, $\delta_{uv} = 0.3$ for pairs involving a metal and 0.5 otherwise. These thresholds are independent of τ_{iso} . [Figure 2](#) provides a concrete example: changing the atom correspondence changes which pairs gain or lose bond order, even when the total event counts agree (A and C).

The decoder returns one bijection witness per distinct signed event class, identifying patterns related by exact source permutations that preserve all thresholded bond-change tests. Matching-preserving shuffles can therefore yield different event classes. The decoder first certifies families whose assignments all yield equivalent events and excludes families outside the requested event window. It then encodes the remaining family constraints symbolically in Z3 (5). Each solution supplies a witness and its event pattern; excluding that entire event class allows the next distinct pattern to be found. This avoids enumerating individual bijections or permutation groups.

For a window $E \leq K$, decoding is complete when every retained family is certified exhausted or excluded. A timeout leaves completeness unresolved. This certificate concerns the saved search output within the window; it does not establish exhaustive exploration of all chemically possible mappings. The default deduplication preserves families that contain different event counts, as illustrated in the heavy-atom projection in [Figure 1c](#).

2.6 Alternative correspondences and chemical interpretation

An atom mapping summarizes the net correspondence resulting from a chemical process without specifying its intermediates or elementary steps. When that pathway is unknown, endpoint structures alone may not determine a unique correspondence. GRAFT therefore retains alternative bond-event patterns and the correlated shuffles within each, exposing one witness per event class together with its supporting families and conditional symmetry queries. Distinct patterns can have equal event counts, and minimizing the count need not recover the reference correspondence.

[Figure 2](#) illustrates both points with Golden case 9. GRAFT recovers the annotated correspondence, in which the alcohol oxygen enters the product ring and the ketone oxygen leaves in water (A). Another candidate exchanges these oxygen origins (B); a third also exchanges two carbon assignments (C). Candidate B has fewer total events than the reference, while A and C have equal totals but different correspondences. Thus, selecting either one bijection or only the lowest event count discards information about the alternatives. These are selected witnesses from the search, not an exhaustive decoded set or validated reaction pathways.

Golden case 9: the reference is one of several recovered mappings

GRAFT search · 1 seed (default) + cut sweep · cap 2,000 · iso tolerance 1 · event threshold 0.5

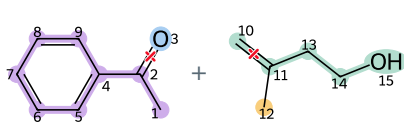
● ● ● Saved fragment groups

—×— Break / weaken

—>— Form / strengthen

A Ground truth (Golden reference) — recovered by GRAFT

5 heavy-atom changes

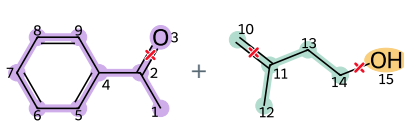


Reference: alcohol O₁₅ enters the ring; ketone O₃ leaves in water.

9 total events = 5 heavy + 4 involving H

B GRAFT alternative — different oxygen retained

6 heavy-atom changes

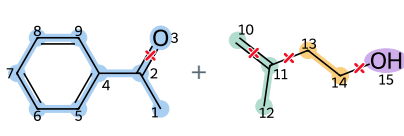


Alternative: ketone O₃ enters the ring; alcohol O₁₅ leaves in water.

8 total events = 6 heavy + 2 involving H

C GRAFT alternative — oxygen and carbon correspondence change

7 heavy-atom changes



Alternative: O₃ enters the ring, with a different assignment of C₁₃ and C₁₄.

9 total events = 7 heavy + 2 involving H

Retaining alternatives matters: a lower total event count (B: 8) need not select the annotated mapping (A: 9).

Ground truth refers to Golden's heavy-atom annotation. H identities are unannotated; totals use each GRAFT witness.

Numbers are original Golden labels. H atoms are implicit. Selected witnesses do not imply exhaustive or minimum-event decoding.

Figure 2 A recovered reference and alternative correspondences from Golden case 9. A matches all 15 annotated heavy-atom pairs. B exchanges the oxygen origins; C also exchanges C₁₃/C₁₄. Pastel colors identify matched fragments. Black lines show endpoint bonds; red × marks indicate cleavage or weakening, and green inward arrows indicate formation or strengthening. H atoms are implicit in the drawing; event totals include H. Each witness minimizes H-involving changes for its fixed heavy-atom correspondence. “Ground truth” denotes the dataset annotation, not a verified pathway. The three witnesses come from one-seed GRAFT sweep search at cap 2,000 and $\tau_{\text{iso}} = 1$, without reference labels; events use threshold 0.5.

3 Evaluation

3.1 Datasets and comparison protocol

We evaluate reference recovery on all 1,851 Golden reactions (8) and bond-event alternatives on 140 coordinate/WBO endpoint pairs. Golden inputs use formal bond orders and explicit H, with reference labels removed before search. Recovery requires the annotated heavy-atom correspondence, including unmatched atoms, modulo endpoint chemical symmetries preserving charge, isotope, H count, and stereochemistry. Failures and unresolved cases remain in the denominator. This measures reference inclusion among returned alternatives, not their ranking.

LocalMapper provides the prior accuracy-SOTA baseline (2). Its published Golden accuracy was 89.8% on 1,758 retained reactions with manual calibration. SLAP reported 86.9% on that dataset version, counting reference inclusion among its candidates (7). These literature scores are separate from our strict re-evaluation. We additionally apply the cut sweep introduced here to SLAP; “SLAP sweep” denotes this extension, not a

Table 1 Evaluation configurations. Both use $b = 0.2$, $\tau_{\text{iso}} = 1$, explicit H, and uncut plus single-edge searches. Default GRAFT uses one seed per cut, the sweep. Golden measures the sweep-search stage; no-sweep controls retain only the uncut graph. A seed setting counts source-atom orderings per cut.

Setting	Golden	Coordinate/WBO
Reactions	1,851	140
Search directions	Bidirectional	Reactant to product
Seed orderings	1 (default), 2, 3, 10	1 (default)
Branch cap	100 (default)	100 (default), 2,000
Evaluation	Reference-family recovery	Signed event-class coverage

published SLAP score. It is our strongest evaluated prior-method baseline for alternative recovery. Released RXNMapper, Chython, Indigo, and RDT provide further comparisons (11; 3; 6; 10).

The coordinate collection contains 18–137 explicit atoms per endpoint, including organometallic and multi-component structures. It was inspected during development and has no verified atom mappings: it is not a mapping-accuracy benchmark. We compare the lowest event count found and the patterns achieving that count. GRAFT uses supplied WBO matrices; SLAP uses XYZ-derived binary graphs. Returned SLAP heavy-atom assignments receive hydrogen refinement and are then scored on the same WBO matrices as GRAFT using Equation (5). The comparison covers those witnesses, not all possible SLAP hydrogen assignments.

Table 1 lists the evaluated settings. Search does not use reference or comparator correspondences. Full configurations, implementation versions, and reproduction scripts are provided in the repository (Appendix A).

3.2 Golden reference recovery and search cost

Default GRAFT recovers 99.08% of Golden references, compared with 97.03% for SLAP sweep (Table 2 and figure 3). The paired comparison includes 43 references recovered only by GRAFT and 5 only by SLAP. LocalMapper is the strongest evaluated method returning one bijection. Additional GRAFT seeds yield modest recovery gains at increasing search cost; two seeds do not improve recovery over the default.

The cut sweep is central to this result: it adds 345 reference recoveries for GRAFT and 135 for SLAP. Without the sweep, one-seed GRAFT has lower coverage than bidirectional SLAP. This supports varying the growth constraints to expose alternatives excluded by greedy fragment extension (Section 2.2); symmetry expansion alone cannot restore discarded placements.

Smaller-first grows fragments from the reaction side with fewer explicit atoms (including H) and matches into the other side; larger-first reverses these roles. Bidirectional search combines both outputs. It adds 31 recoveries over smaller-first alone, demonstrating that the two orientations retain complementary alternatives (Table 3).

Default GRAFT has a lower mean and median search cost than SLAP sweep, but a higher 95th percentile. Two seeds approximately double the mean cost without improving recovery here. Golden timings cover search, excluding reference verification and separate bond-event decoding. The two timing tables use different experiments and should be compared within their stated CPU and reaction sets.

Increasing the branch cap from 100 to 2,000 yields no additional swept recoveries among cases resolved at both caps, while increasing cost and unresolved outcomes at higher seed counts (Table A3). The uncut search gains 19 confirmed recoveries. Thus, a larger branch budget does not substitute for changing the growth conditions.

The default-comparator rows in Table 2 use released interfaces without prediction repair or reference-guided reranking. Binary SLAP is retained as the stronger default configuration. The other methods return at most one explicit bijection per reaction, whereas SLAP returns multiple candidates. Invalid endpoint chemistry and duplicate atom labels count as failures; the repository records their frequencies. These results describe the evaluated interfaces and scoring rule, including its difference from RDT’s chemical-change criterion.

Table 2 Golden reference recovery and runtime. Recovery uses all 1,851 primary-evaluation records. GRAFT retains compressed families; SLAP returns explicit candidates; other methods return one bijection over their matched atoms. GRAFT uses cap 100 and bidirectional search. Bidirectional SLAP combines both directions and both binary/weighted modes; the default SLAP row uses binary only. Times are mean CPU seconds per reaction, excluding separate bond-event decoding. Unmarked times use the same 1,403 completed reactions on an AMD EPYC 9J14. *a*: archived completed calls (1,718–1,851 per method), CPU model unrecorded; these times are not directly comparable with unmarked times. †: LocalMapper, prior accuracy SOTA (2024). ‡: SLAP with our sweep, strongest evaluated prior-method recovery baseline.

Method	Search setting	Output	Reference recovered	Mean CPU s
GRAFT	1 seed, sweep (default)	Families	1,834 (99.08%)	0.963
GRAFT	2 seeds, sweep	Families	1,834 (99.08%)	1.619
GRAFT	3 seeds, sweep	Families	1,837 (99.24%)	2.265
GRAFT	10 seeds, sweep	Families	1,840 (99.41%)	6.993
GRAFT	1 seed, no sweep	Families	1,489 (80.44%)	0.040
SLAP, bidirectional	No sweep	Candidates	1,661 (89.74%)	0.105
SLAP, bidirectional‡	Sweep	Candidates	1,796 (97.03%)	2.793
SLAP, binary	Default, no sweep	Candidates	1,591 (85.95%)	0.075 ^a
RXNMapper	Default, no sweep	One bijection	1,547 (83.58%)	0.056 ^a
LocalMapper†	Default, no sweep	One bijection	1,605 (86.71%)	0.294 ^a
Chython	Default, no sweep	One bijection	1,561 (84.33%)	0.612 ^a
Indigo	Default, no sweep	One bijection	692 (37.39%)	0.184 ^a
RDT	Default, no sweep	One bijection	1,030 (55.65%)	1.021 ^a

Table 3 Recovery and runtime by search orientation. All rows use the sweep. Recovery uses all 1,851 Golden records; CPU statistics are seconds per reaction over the same 1,821 completed reactions (Apple M4 Pro CPU). Smaller-first grows from the side with fewer explicit atoms, including H; larger-first reverses the source and target. Equal-size ties use stored order and its reverse. Bidirectional output combines both searches, and its CPU cost is their sum. SLAP combines binary and weighted modes. CPU/SLAP compares means within each orientation. Timing scopes are specified in [Appendix A](#).

Method	Orientation	Recovered	Mean	Median	95th pct.	CPU/SLAP
GRAFT, 1 seed (default)	Smaller-first	1,803 (97.41%)	1.141	0.263	5.289	0.34
	Larger-first	1,779 (96.11%)	1.377	0.348	6.401	0.92
	Bidirectional	1,834 (99.08%)	2.518	0.719	12.913	0.52
GRAFT, 2 seeds	Smaller-first	1,815 (98.06%)	2.232	0.416	10.635	0.67
	Larger-first	1,805 (97.51%)	2.721	0.578	14.064	1.83
	Bidirectional	1,834 (99.08%)	4.953	1.250	26.350	1.03
SLAP sweep (prior baseline)	Smaller-first	1,775 (95.89%)	3.329	0.422	4.752	1.00
	Larger-first	1,770 (95.62%)	1.490	0.455	4.568	1.00
	Bidirectional	1,796 (97.03%)	4.819	0.881	11.048	1.00

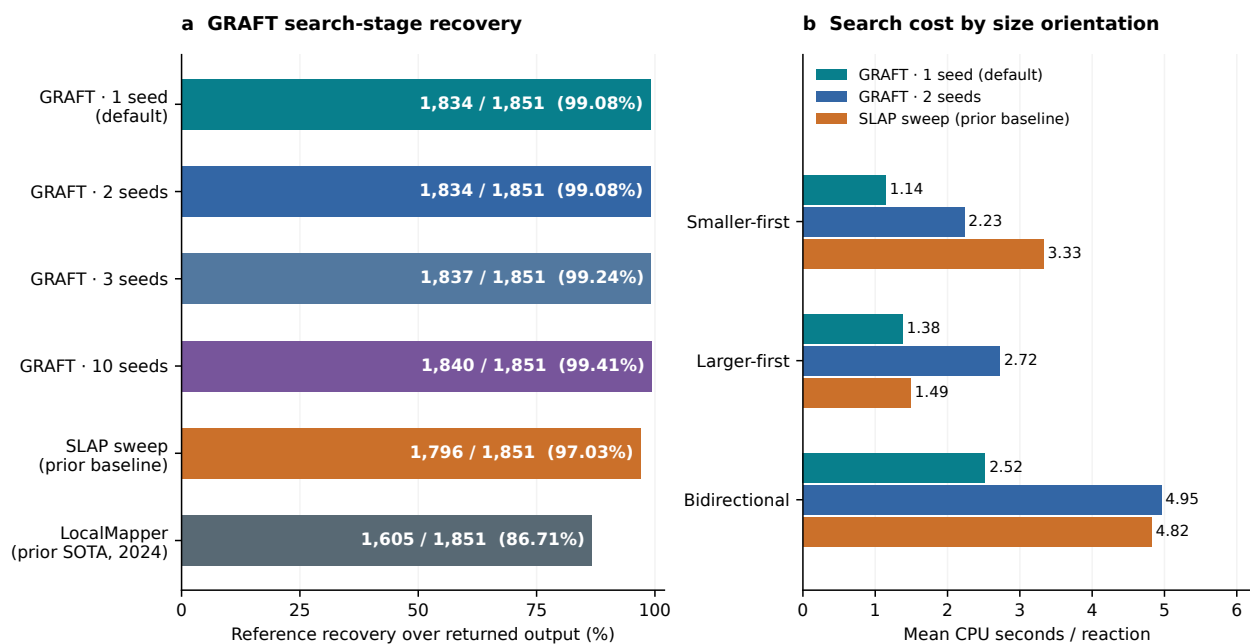


Figure 3 Golden reference recovery and search cost. (a) Reference recovery over all 1,851 reactions. GRAFT and SLAP use our cut sweep; LocalMapper is the prior accuracy-SOTA baseline and returns one bijection. All scores use our evaluator. (b) Mean search CPU over the same 1,821 completed reactions, by orientation. Bond-event decoding is excluded. Full comparisons and timing scopes are in [Tables 2 and 3](#).

3.3 Minimum bond-event comparison on 140 coordinate reactions

We use one-seed GRAFT sweep search at cap 2,000, which returns complete mappings for all 140 reactions. Cap 100 leaves two reactions incomplete (cases 123 and 125). The larger cap is therefore used for this comparison; the library default remains 100.

GRAFT matches or lowers the minimum event count found by both native SLAP and SLAP sweep in every reaction (Table 4 and figure 4). This does not imply recovery of every alternative: at equal minima, six SLAP-sweep patterns remain absent across four reactions. An equal count can describe different changed bonds, and a lower count alone does not establish a more plausible transformation.

Table 4 Minimum bond-event comparison on 140 reactions. GRAFT uses one-seed sweep search at cap 2,000. The first three columns count reactions by GRAFT’s minimum event count relative to the comparator. The final column gives comparator minimum-event patterns recovered by GRAFT among reactions with equal minima. Counts refer to returned candidates, not proven global optima.

Compared with	GRAFT fewer	Equal	GRAFT more	Patterns at equal minima
SLAP sweep	4	136	0	158/164
Native SLAP	15	125	0	138/140

To compare alternatives above GRAFT’s own minima, we decode through one event above the higher of the two SLAP minima for each reaction. GRAFT retains 162 of 168 SLAP-sweep minimum-event patterns and 151 of 160 native-SLAP patterns within these windows. The output contains 300 distinct patterns, including 166 at GRAFT’s own minima. Every saved family is decoded or certified within its window; this establishes completeness for the retained families, not exhaustive structural search.

Table 5 Stage CPU time per reaction on the 140 coordinate reactions. Measurements use an Apple M4 Pro CPU. GRAFT uses one-seed sweep search at cap 2,000. These stage costs exclude GRAFT decoding and do not establish an end-to-end speed ranking. SLAP refinement covers its returned witnesses. Timing boundaries are specified in Appendix A.

Recorded stage	Mean CPU s	Median CPU s	95th pct. CPU s
GRAFT sweep search	1.047	0.314	3.481
SLAP sweep mapping	1.337	0.546	5.919
SLAP H refinement / initial scoring	0.053	0.011	0.207

The stage timings in Table 5 cover different outputs: GRAFT search retains families for subsequent decoding, while SLAP refinement operates on returned witnesses. They therefore do not establish an end-to-end speed ranking.

A fresh run of the final implementation completed search and event decoding for 140/140 coordinate reactions. On this full set, the full pipeline takes a mean 6.07 CPU seconds per reaction (median 0.81; 95th percentile 25.47). Decoding accounts for a mean 4.65 CPU seconds (median 0.29). These measurements include search, final-family construction, decoding, and candidate output; they are separate from the search-only comparison above (Table A4).

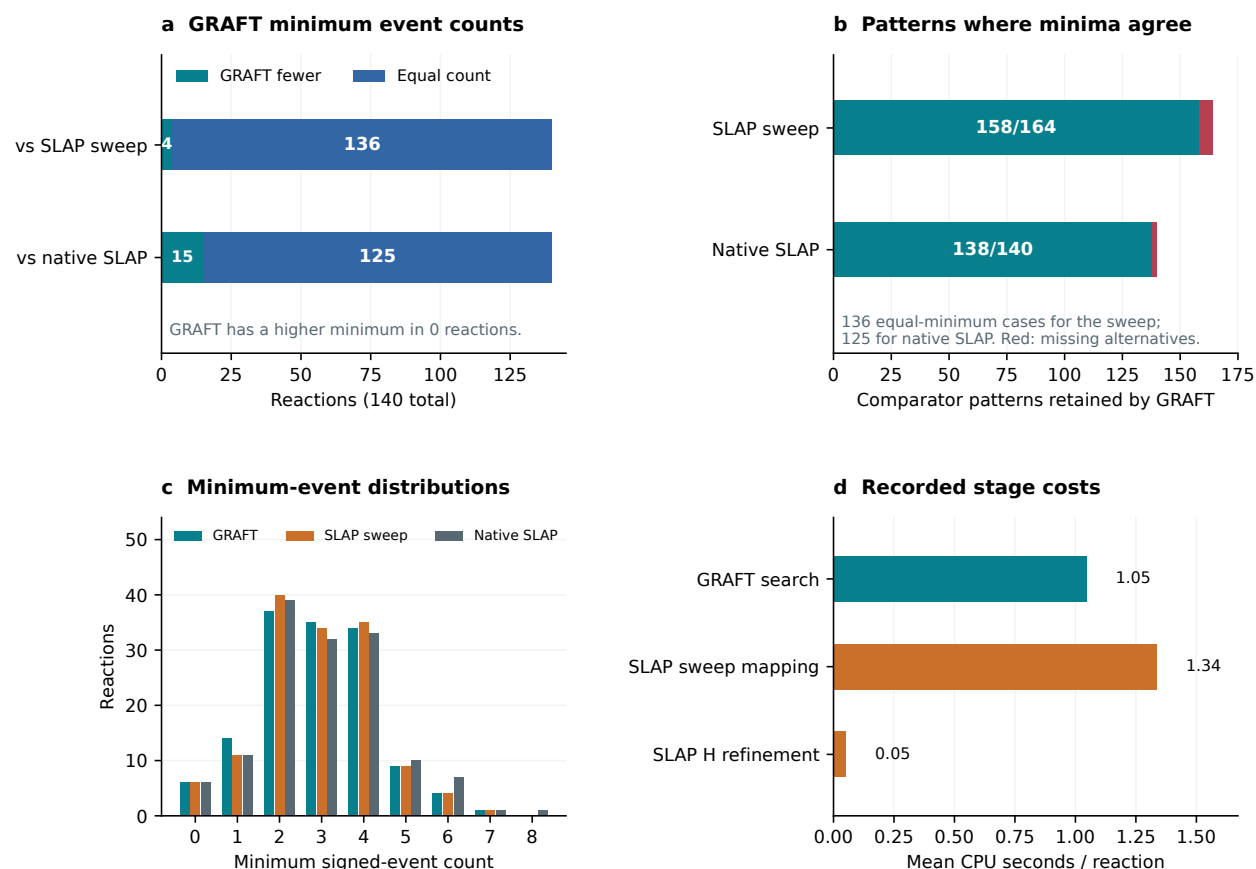


Figure 4 Bond-event comparison on 140 coordinate reactions. GRAFT uses one seed, the sweep, and cap 2,000. (a) Reactions with fewer or equal GRAFT events than each SLAP baseline. (b) Comparator minimum-event patterns recovered by GRAFT at equal minima. (c) Minimum-event count distributions. (d) Mean CPU for search and SLAP refinement, excluding GRAFT decoding (Table 5). Atom mappings are unverified; this compares bond events, not mapping accuracy.

4 Discussion

4.1 Implications for pathway and transition-state analysis

Different event patterns can suggest different reaction centers, while allowed joint atom shuffles within a pattern can change the product ordering used for endpoint interpolation. A transition-state workflow can query these assignments against GRAFT’s family constraints and test several initial paths instead of committing to one bijection. Atomic-index assignment is also a recognized obstacle in reaction-pathway search (4).

These alternatives are starting hypotheses, not identified mechanisms. One net correspondence can admit several intermediate sequences, and different correspondences need not be chemically plausible. Energies, reaction conditions, stereochemistry, and isotope tracing provide information absent from endpoint connectivity. We do not evaluate transition-state search performance here.

Alternative enumeration is not unique to GRAFT or SLAP; for example, an Ising-computing approach enumerates optimal mappings and reduces them by molecular symmetry (1). GRAFT’s contribution is to retain conditional fragment families and expose their distinct event patterns and allowed shuffles without expanding all bijections.

4.2 Coverage and remaining search limitations

An audit of all eleven ten-order Golden misses separates unmatched-atom decisions from fragment-placement failures (Table A2). In four reactions, every retained family maps exactly one more heavy-atom pair than the reference. Such a family cannot reproduce the reference’s unmatched-atom pattern through symmetry shuffling: permutations preserve the number of assigned pairs. The remaining seven lack the required correspondence within their saved families. Re-ranking or expanding those families into explicit bijections cannot supply a missing structural alternative.

Two traced examples show how greedy extension excludes a representable reference. One reference exchanges adjacent labels in an otherwise unchanged pyridine ring; another exchanges the carbonyl-oxygen attachments of oxalic acid across its retained carbons. Ordinary growth preserves bonds that these correspondences require separating. Uncapped trajectories still miss both references, while reference-constrained fragment choices establish their representability. In the oxalic-acid example, a reference-selected three-edge relaxation also exposes the reference to ordinary matching; the single-edge sweep does not contain that cut set. These are feasibility diagnostics, not additional benchmark recoveries. For the other five correspondence misses, the audit does not isolate cap truncation from unsampled growth choices. They motivate selective alternatives to successful growth extensions and explicit choices about leaving atoms unmatched, rather than treating larger caps or more symmetry expansion as sufficient remedies.

Only one of the eleven misses is recovered by any tested comparator configuration, including the SLAP sweep. The shared failures warrant inspection of the supplied correspondences as well as search improvements; they do not establish annotation errors. All eleven remain in the reported denominator. Independent evaluation on annotated coordinate reactions is needed to assess generalization beyond the development collection.

Code and data

Code and reproduction instructions are available at <https://github.com/yunhzou/GRAF-AAM>, branch `main`. The repository guide `manuscript/REPRODUCIBILITY.md` links full benchmark configurations, software versions, per-case results, and figure-generation scripts. Golden is distributed by its original authors (8).

Acknowledgements

We thank Kjell Jorner and Robert Pollice for helpful discussions.

References

- [1] Mohammad Ali, Yuta Mizuno, Seiji Akiyama, Yuuya Nagata, and Tamiki Komatsuzaki. Enumeration approach to atom-to-atom mapping accelerated by Ising computing. *Journal of Chemical Information and Modeling*, 65(4):1901–1910, 2025. doi: 10.1021/acs.jcim.4c01871.
- [2] Shuan Chen, Sunggi An, Ramil Babazade, and Yousung Jung. Precise atom-to-atom mapping for organic reactions via human-in-the-loop machine learning. *Nature Communications*, 15:2250, 2024. doi: 10.1038/s41467-024-46364-y.
- [3] Chython contributors. Chython: molecular and reaction processing toolkit. <https://github.com/chython/chython>. Evaluated version 2.18 with chython-rxnmap 2.0; model hash archived.
- [4] Deb Sankar De, Marco Krummenacher, Bastian Schaefer, and Stefan Goedecker. Finding reaction pathways with optimal atomic index mappings. *Physical Review Letters*, 123:206102, 2019. doi: 10.1103/PhysRevLett.123.206102.
- [5] Leonardo de Moura and Nikolaj Bjørner. Z3: An efficient SMT solver. In *Tools and Algorithms for the Construction and Analysis of Systems*, pages 337–340. Springer, 2008. doi: 10.1007/978-3-540-78800-3_24.
- [6] EPAM Systems. Indigo cheminformatics toolkit. <https://github.com/epam/Indigo>. Evaluated version 1.46.0.

- [7] Shin-ichi Koda and Shinji Saito. General and scalable atom-to-atom mapping via Weisfeiler–Lehman-like approximate graph matching. *ChemRxiv*, 2025. doi: 10.26434/chemrxiv-2025-hthwn. Preprint, version 1, 25 November 2025.
- [8] Arkadii Lin, Natalia Dyubankova, Timur I. Madzhidov, Ramil I. Nugmanov, Jonas Verhoeven, Timur R. Gimadiev, Valentina A. Afonina, Zarina Ibragimova, Assima Rakhimbekova, Pavel Sidorov, Andrei Gedich, Rail Suleymanov, Ravil Mukhametgaleev, Joerg Wegner, Hugo Ceulemans, and Alexandre Varnek. Atom-to-atom mapping: A benchmarking study of popular mapping algorithms and consensus strategies. *Molecular Informatics*, 41(4):2100138, 2022. doi: 10.1002/minf.202100138.
- [9] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60: 94–112, 2014. doi: 10.1016/j.jsc.2013.09.003.
- [10] S. A. Rahman, G. Torrance, L. Baldacci, S. Martínez Cuesta, F. Fenninger, N. Gopal, S. Choudhary, J. W. May, G. L. Holliday, C. Steinbeck, and J. M. Thornton. Reaction decoder tool (RDT): extracting features from chemical reactions. *Bioinformatics*, 32(13):2065–2066, 2016. doi: 10.1093/bioinformatics/btw096.
- [11] Philippe Schwaller, Benjamin Hoover, Jean-Louis Reymond, Hendrik Strobelt, and Teodoro Laino. Extraction of organic chemistry grammar from unsupervised learning of chemical reactions. *Science Advances*, 7(15):eabe4166, 2021. doi: 10.1126/sciadv.abe4166.

A Evaluation details

A.1 Reproducibility and comparison scope

The repository’s `manuscript/REPRODUCIBILITY.md` links the benchmark scripts, pinned software and input versions, full configurations, and per-case records. [Table 1](#) gives the settings needed to interpret the comparisons. The notebook `docs/AAM_SIMPLE.ipynb` demonstrates matching, unique event candidates, and conditional symmetry queries through the Python API.

For the coordinate comparison, GRAFT uses supplied WBO matrices and SLAP uses the original XYZ-derived binary graphs. SLAP’s returned heavy-atom assignments receive hydrogen refinement under its original bond-change objective, then are rescored on the WBO matrices. Thus, the comparison uses a common event score but different input representations. It covers the returned SLAP witnesses, not an exhaustive expansion of its hydrogen label families. The WBO-generation method is not recorded in the available metadata, so regeneration of these matrices from molecular structures is not established.

The paired search timings use an Apple M4 Pro CPU. Averages use 1,821 common completed reactions. Reference verification and interrupted work are excluded. GRAFT includes checkpoint I/O; SLAP excludes I/O. Higher-seed runs measured on different CPUs are excluded from this paired comparison. For the coordinate stage comparison, GRAFT search includes checkpoint I/O; SLAP mapping excludes input preparation and export, and its hydrogen refinement includes I/O. Shared event rescoring and failed or replaced attempts are excluded. Golden GRAFT timing includes symmetry finalization; SLAP includes graph construction, mapping, and export. In the primary Golden evaluation, uncut and swept SLAP have one and three unresolved cases, respectively; all GRAFT verdicts are resolved.

A.2 Default comparator protocol

Released comparators receive the same label-free Golden inputs, without prediction repair or reference-guided ranking. Invalid outputs and failures remain in the denominator. SLAP is evaluated over all returned candidates, using binary mode for the default comparison and both modes for bidirectional comparisons. The repository records the released models, versions, and adapter settings.

A.3 Decoding scope

For each coordinate reaction, the decoding limit is one event above the larger of the native-SLAP and SLAP-sweep minima. All 120,052 retained families are exhausted within these limits, without a class-count cap. Higher-event patterns are outside this completeness claim. An event class records the signed atom-pair changes, modulo source permutations preserving every thresholded bond-change test against the opposite endpoint. Each class has a concrete mapping witness.

Table A1 Complete final decoding on all 140 reactions. Every retained family is certified within the listed per-case event limit.

Maximum events	1	2	3	4	5	6	7	8	9
Reactions (all complete)	6	11	39	32	33	10	7	1	1

A.4 Preservation of mapping families under flattening

Flattening retains each terminal representative, its required fragment constraints, and its allowed action sequence after applying prefix locks. Disjoint actions commute. Adjacent nested groups satisfy $HG = GH = G$ when $H \leq G$, with containment checked by exact generator membership. Other overlapping actions retain their order. Identical records are shared; branches with the same unordered fragment pairs retain the union of their family records. The admissible mapping union is therefore unchanged. A shared fragment partition or automorphism group alone does not establish family equality.

The decoder certifies a family through event invariance, a valid lower bound excluding the requested window, or unsatisfiability after excluding all discovered event classes. An exhausted budget remains incomplete. This separates lossless representation, complete decoding within a window, and exhaustive structural search.

A.5 Golden nonrecoveries

Table A2 The eleven Golden nonrecoveries. Case indices are zero-based. Retained pair counts cover every terminal in both directions and every saved cut at ten seed orderings and cap 100. The four cardinality mismatches concern exact reference relations, including unmatched atoms.

Case	Reference pairs	Retained pairs	Observed mismatch
7	29	30	One extra matched pair
19	23	23	Atom pairing excluded by orbit bounds
590	31	31	Atom pairing excluded by orbit bounds
871	20	20	Atom pairing excluded by orbit bounds
986	16	16	Atom pairing excluded by orbit bounds
1228	12	12	Atom pairing excluded by orbit bounds
1285	14	14	Joint C/O correspondence excluded
1358	36	37	One extra matched pair
1377	22	23	One extra matched pair
1475	18	19	One extra matched pair
1553	20	20	Atom pairing excluded by orbit bounds

Independent reconstruction agrees with the supplied references in all eleven cases. Four require a different number of matched pairs; seven require correspondences absent from the retained families. All have at least one capped direction, but this does not identify the cap as the cause. The uncapped traces discussed in the main text demonstrate greedy-growth exclusions in two cases. Reference-directed feasibility checks are diagnostic and do not contribute to benchmark recovery. Per-case certificates and traces are linked in the repository.

A.6 Branch-cap and CPU-time ablation on Golden

We reran the twelve configurations in [Table A3](#) on all 1,851 Golden reactions using an AMD EPYC 9J14 CPU. Both search directions are unioned. SLAP unions its binary and weighted modes. Search and reference verification have separate fixed resource budgets. The recorded cap-100 recovery counts reproduce the main evaluation. At cap 2,000, unresolved counts increase with seed count. Among reactions resolved at both caps, no swept configuration gains or loses a reference recovery.

The ten-seed cap-2,000 run leaves five references recovered at cap 100 unresolved, accounting for its lower confirmed count. The bounded SLAP sweep confirms 1,794 references here, compared with 1,796 in the main evaluation: two previously recovered cases are unresolved under this study’s limits. These outcomes are reported separately from verified exclusions. The paired timing cohort excludes 448 reactions because at least one configuration does not finish; the all-attempt cost column exposes the additional computational cost on those difficult reactions. Source pins, individual outcomes, directional costs, and reproduction scripts are provided with the code.

Table A3 Golden branch-cap ablation with paired SLAP baselines. Recovery counts confirmed reference inclusion over all 1,851 records; unknown cases remain in the denominator. CPU values are seconds per reaction. Paired CPU uses the same 1,403 reactions whose searches finish under every configuration. Spent CPU averages recorded search cost over all 1,851 selected attempts, including interrupted calls; it is not completed-search latency. Both timings exclude reference verification and initial imports. Superseded attempts are excluded. Cap 100 with a one-seed sweep is the default.

Method	Cap	Seeds	Recovered (%)	Unknown	Paired CPU	Spent CPU
GRAFT uncut	100	1	1,489 (80.44)	0	0.040	0.085
GRAFT sweep (default)	100	1	1,834 (99.08)	0	0.963	3.873
GRAFT sweep	100	2	1,834 (99.08)	0	1.619	7.260
GRAFT sweep	100	3	1,837 (99.24)	0	2.265	10.556
GRAFT sweep	100	10	1,840 (99.41)	0	6.993	33.211
GRAFT uncut	2,000	1	1,508 (81.47)	5	0.086	0.517
GRAFT sweep	2,000	1	1,834 (99.08)	2	4.317	30.055
GRAFT sweep	2,000	2	1,834 (99.08)	4	8.283	48.549
GRAFT sweep	2,000	3	1,837 (99.24)	6	12.201	61.890
GRAFT sweep	2,000	10	1,835 (99.14)	12	43.843	113.777
SLAP uncut	–	–	1,661 (89.74)	1	0.105	0.659
SLAP sweep	–	–	1,794 (96.92)	5	2.793	9.895

A.7 Fresh end-to-end timing on coordinate reactions

We measured one-seed, cap-2,000 cut-sweep search followed by the public event decoder on the Apple M4 Pro CPU. CPU times are reported per reaction. The event thresholds and windows match the bond-event comparison, with no class-count cap. All 140 completed reactions reproduce the archived event-class sets and family counts. Median end-to-end wall time is 0.96 seconds. Timing includes input loading, checkpoint I/O, final-family construction, decoding, and candidate serialization; initial imports and reference-set verification are excluded. All 140 reactions finish within the five-minute watchdog. Detailed timing records and reproduction instructions are provided with the code.

Table A4 GRAFT search plus bond-event decoding. CPU seconds per completed reaction in the fresh 140-reaction pass. The final row measures the whole pipeline; medians and percentiles of individual stages are not additive. Completion coverage and timing scope are stated above.

Stage	Mean CPU s	Median CPU s	95th pct. CPU s
GRAFT sweep search	1.257	0.300	3.468
Final-family catalogue	0.156	0.019	0.595
Bond-event decoding	4.647	0.288	24.361
Candidate serialization	0.005	0.001	0.015
End to end	6.073	0.812	25.470